



Metabolic activity and pathway study of emerging contaminants biodegradation using a photo-bioelectrochemical system: a review

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Abstract

Emerging contaminants and their intermediate residue pose significant risks to human health and ecosystems due to their persistence in water and soil. The review explores the potential of photo-bioelectrochemical systems as a dual-function approach for the degradation of emerging contaminants while simultaneously generating bioenergy. Photo-bioelectrochemical systems integrate microbial metabolism with photocatalytic processes, utilizing solar energy to enhance electron transfer and promote the breakdown of toxic compounds. Recently, this technology has demonstrated improved removal rates of various emerging contaminants, including antibiotics and neonicotinoid pesticides, while producing valuable by-products such as hydrogen and methane through optimized operational conditions. Although studies have focused mainly on pharmaceuticals and dyes, applying these systems for pesticides remains underexplored. Algae-assisted photo-bioelectrochemical systems have been observed frequently in recent studies. This review synthesizes current research on the metabolic activity and biochemical pathways involved in the biodegradation of contaminants within photo-bioelectrochemical systems. It highlights the efficiency of these systems in converting harmful substances into less toxic by-products. It addresses challenges such as slow degradation rates compared to chemical methods and the need for enhanced extracellular electron transfer capabilities among microorganisms. Through optimized operational conditions, this paper aims to provide a comprehensive understanding of photo-bioelectrochemical systems' performance and to propose future research directions for optimizing these systems in real-world applications. In conclusion, this work underscores the promise of photo-bioelectrochemical systems as a sustainable solution for wastewater treatment and environmental remediation.

Keywords Photocatalyst · Emerging contaminants · Algae · Bioelectrochemical system · Biocatalyst · Pathway

Introduction

Emerging contaminants (ECs) are a group of different contaminants such as chemical compounds and microorganisms frequently found in pharmaceutical industries, hospital wastewater, solid waste leachate, and influent and effluent of municipal wastewater treatment plants (WWTPs), and their sewage sludge (He et al. 2024). ECs can harm human health and ecosystem safety due to their persistence in water resources and soil (Moghiseh et al. 2023). Studies have reported high concentrations of pharmaceuticals in aquatic environments; for example, the maximum concentrations were 17.17 ng/L in the drinking water of China, 1820 ng/L in

the groundwater of Taiwan, and 7.24 ng/L in the river water of Bangladesh, highlighting the inefficacy of conventional treatment processes in WWTPs. Patel et al. (2019) confirmed the presence of pharmaceutical contaminants throughout the environment and highlighted issues such as high annual and operating costs, unsuitable removal, and high toxic sludge production associated with various removal methods (Patel et al. 2019). Antibiotic, anti-inflammatory, and analgesic compound residues are commonly found in wastewater and subsequently in offshore seawater. Penicillin, ciprofloxacin, sulfamethoxazole, cefixime, azithromycin, and tetracycline are frequently detected in both the influent and effluent of wastewater treatment plants in Oman and European nations (Moghiseh et al. 2023). In addition, pesticides classified as pharmaceuticals are found in water sources. Imidacloprid, a neonicotinoid pesticide, is highly soluble (610 mg L^{-1} at 20°C and pH 7) and stable. It has often been detected in groundwater, soil, wetlands, and even drinking water, as

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it enters the environment through spraying and leaching, with a potential index of 3.76. In water obtained from the drinking water treatment system at the University of Iowa in the USA, imidacloprid and other neonicotinoid pesticide concentrations reached as high as 57 ng L^{-1} . In Canada, imidacloprid and clothianidin, another neonicotinoid pesticide, were detected at concentrations of up to $0.5 \text{ } \mu\text{g L}^{-1}$ and $0.3 \text{ } \mu\text{g L}^{-1}$, respectively, in drinking water sources. These levels exceed the standard guideline value of $0.1 \text{ } \mu\text{g L}^{-1}$ for individual pesticides in Canada guidelines (Deng et al. 2022b). Dinotefuran, also a neonicotinoid pesticide, has a solubility of $39,830 \text{ mg L}^{-1}$ in water and is frequently found in the environment. It is anticipated to reach a maximum concentration of $154 \text{ } \mu\text{g L}^{-1}$ in groundwater (Yin et al. 2023). Therefore, it is necessary to treat ECs from aquatic environments. Bioelectrochemical systems (BESs) have garnered substantial interest due to their unique ability to combine microbial metabolism with electrochemical processes. This combination enables the simultaneous treatment of wastewater and recovery of bioenergy, offering a promising solution for sustainable wastewater management and energy production (Yi et al. 2022). BESs can enhance the detoxification and degradation of toxic organic pollutants while promoting the conversion of organic and inorganic contaminants into less harmful forms and valuable products. A unique redox environment created by the cathode and anode in BESs provides favorable conditions for electroactive microorganisms to carry out these processes (Li et al. 2020a, b). Researchers have shown significant interest in the use of BESs for the remediation of different persistent organic pollutants, including pesticides, phenolic substances, dyes, brominated flame retardants, antibiotics, organophosphate esters, and aromatic ring compounds (Rafieenia et al. 2022). Despite these advancements, traditional BESs have a slower degradation rate for organic pollutants than chemical or electrochemical methods, which could hinder their practical use, especially for ECs that are toxic to microbial communities at high concentrations. The effectiveness of degradation in BESs is significantly affected by the capacity of microorganisms to transfer electrons outside their cells within the system. While BESs have achieved improved degradation efficiency through the facilitation of extracellular electron transfer (EET), the comparatively low theoretical EET efficiency of microorganisms still poses a challenge compared to chemical reaction-based methods. The efficiency of EET is a key factor determining the performance of BESs and their practical applications (Wang et al. 2022).

To overcome this problem, photo-bioelectrochemical systems (PBESs) integrate BESs and photocatalysts or photochemical processes in one reactor, utilizing solar energy (UV and visible) to generate fast bioelectricity and valuable products while simultaneously treating organic wastes such as ECs (Rafieenia et al. 2022; Sun et al. 2017). Light sources,

such as solar energy simulators or LED lights, can improve the performance and electron transfer of bioreactors by decomposing toxic emerging contaminants that are not easily degradable by microbes (Rafieenia et al. 2022). A study revealed the efficient removal of phenol and aniline contaminants, along with their COD reduction at high concentrations, using a combined photo-electrocatalytic (UV-visible) and BES (PEC-BES) process. It also reported the complete removal of 50 mg L^{-1} 2,4,6-trichlorophenol (TCP) within 72 h using the PEC-BES process (Zhang et al. 2019b). Other studies have coupled PBESs with additional wastewater treatment processes to enhance the removal of ECs. For example, a hybrid persulfate-photo-bioelectrochemical (PPBEC) system has been proposed to eliminate ECs such as carbamazepine and clofibric acid from effluent, achieving higher removal rates and lower energy consumption compared to individual processes (Zou et al. 2021). Additionally, photoelectrode materials have been developed to enhance the removal of ECs (e.g., pharmaceuticals) through photosynthetic complexes operating on conductive surfaces, owing to their stability and biocompatibility properties (He et al. 2024; Yehezkeili et al. 2014). Moreover, pesticides can be obliterated, especially at low concentrations, and can convert into low-toxicity by-products, which may be less toxic than the leading product in the algae-assisted BES (Deng et al. 2022a). Additionally, using green algae-based biological photovoltaic cells (BPVs) as biosensors has enabled the detection of pesticide concentrations in water while also generating a high electric current (Buyukharman et al. 2024).

Operational factors such as oxygen levels, pH, and applied voltage are crucial in influencing the performance and degradation mechanisms of PBES and may require optimization that is distinct from other bioreactors. PBESs also provide suitable conditions for microorganisms that frequently produce valuable products such as methane, hydrogen, and hydrogen peroxide.

In recent years, numerous groundbreaking research papers have been published on PBESs. However, most review studies focus on treating recalcitrant contaminants, energy production, microbial communities, and photoelectrode materials in bioreactors. For instance, Rafieenia et al. (2022) reviewed the mechanism of photocatalysis, the configuration of photo-BESs, microbial communities in BESs, and the interactions between microorganisms and photoelectrodes for treating organic recalcitrant contaminants in wastewater. Apollon et al. (2021) reported on plant BESs, highlighting their progress in system structure and electrode materials, as well as their applications in wastewater treatment (Apollon et al. 2021; Rafieenia et al. 2022). The literature indicates that electricity and hydrogen gas can be generated through the synergistic interaction of biofilms, light sources, and electrical currents, all while simultaneously

remediating organic and inorganic contaminants (Yi et al. 2022; Luo et al. 2017).

This review aims to thoroughly examine and integrate the current literature regarding the metabolic processes and biochemical pathways involved in the biodegradation of ECs using PBESs. The objectives are to evaluate metabolic pathways, assess system efficiency, and propose future research directions. Additionally, to date, there has not been a comprehensive review that considers the key operational conditions in PBESs used for concurrent remediation. These conditions include pH, oxygen levels, applied voltage, light intensity, temperature, conductivity, substrate type, and concentration, as well as semiconductor photocatalysts and reactor designs.

Benefits of PBESs and their recent developments

Integrating photocatalysis with a bioelectrochemical system (BES) can enhance the growth of photoautotrophs (e.g., cyanobacteria and algae) and phototrophs within the biofilm of photo-bioelectrochemical systems (PBESs). This integration primarily utilizes light energy to drive biochemical reactions, enabling simultaneous remediation of electrochemicals from wastewater, energy recovery and generation, hydrogen production, and carbon dioxide reduction (Yi et al. 2022; Xiao et al. 2012). Employing this integrated strategy amplifies the system's performance and sustainability. For example, combining photocatalytic devices with BESs can improve dye removal while concurrently producing electricity (Yi et al. 2022). PBESs can also employ microalgae as biocatalysts to decrease water salinity (Yi et al. 2022). Researchers have developed new BESs that utilize solar light sources and anode photocatalyst materials, such as cobalt phosphate and magnesium hydroxide, to enhance electricity production compared to conventional systems. This enhancement is achieved by facilitating better interactions between light-absorbing materials and electroactive biofilms (Rafieenia et al. 2024). Recent advancements in PBESs include new reactor designs that enhance mass transport, light penetration, and product extraction in photobioreactor systems used for microalgae cultivation to optimize microbial community composition. One example is a PBES reactor integrated with dark fermentation, which has been demonstrated to increase total energy recovery from organic waste (Acién et al. 2017). Moreover, current research on PBES is advancing toward commercialization, focusing on testing new materials and designs to evaluate their scalability, as well as their economic and environmental feasibility. This transition is essential for converting laboratory achievements into practical applications suitable for real-world use. Also, the applications of BES are increasingly broadening beyond energy generation to encompass environmental

remediation, nutrient recovery, and even food safety evaluations using advanced biosensor technologies (Ieropoulos et al. 2024).

The performance of PBES in degrading contaminants is influenced by several factors, including light, device components, and microbial biofilms. To achieve optimal conditions, the start-up process must be carefully managed to promote microbial growth and minimize internal resistance. The main operating conditions are discussed below.

Operational factors

Photocatalytic materials or photocatalysts

The photocatalytic BESs benefit from the increased potential of the cathode, which results from the response of photocatalytic materials, such as TiO_2 , to light. This enhanced potential plays a crucial role in improving the overall efficiency of pollutant degradation by generating reactive oxygen species and participating in oxidation and reduction reactions (Tong et al. 2022; Zhu et al. 2021). These compounds are explained in the following sections. Figure 1 presents the photocatalytic mechanism resulting from the response of semiconductor photocatalytic materials to light (Tong et al. 2022; Zhang et al. 2019a). As shown in Fig. 1, when photocatalysts are exposed to a specific energy of light, electrons in the valence band (VB) of the photocatalyst become excited and move to the conduction band (CB), resulting in the formation of positively charged holes in the valence band. The excited electrons in the conduction band then migrate to the surface of the photocatalyst, where they participate in a reduction reaction by accepting electrons from other chemical species. Meanwhile, the holes in the valence band migrate to the surface of the photocatalyst, where they are involved in oxidation reactions by donating electrons to other chemical species (Tong et al. 2022; Zhang et al.

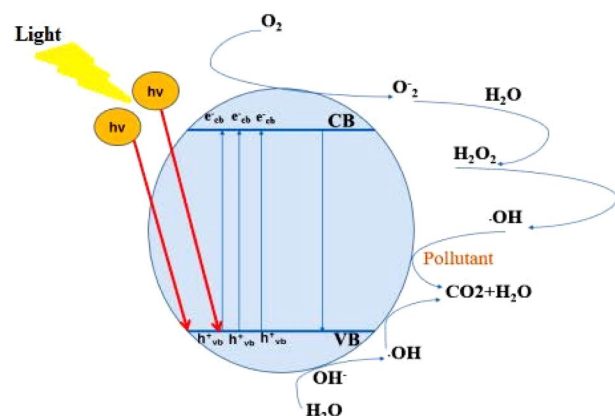


Fig. 1 Illustration of the photocatalytic mechanism at the cathode (Zhang et al. 2019a; Tong et al. 2022)

2019a). This process improves the elimination of ECs such as pharmaceuticals, personal care products, and endocrine disruptors (Li et al. 2024; He et al. 2022).

In a study by Hou et al. (2017a), a photocathode covered with TiO_2 was used in a single-chamber PBES to remove the methyl orange (MO) contaminant. The presence of light and photoelectrodes or photocatalytic material also impacts the microbial community structure and activity. For example, a study discovered significant changes in the microbial community on the cathode biofilm, with enrichment of exoelectrogenic and TCP-degrading microorganisms (Hou et al. 2017a; Zhu et al. 2021; Rafieenia et al. 2022). Photocatalytic materials can enhance microbial activity by creating a conducive environment for microorganisms and facilitating electron transfer between the bioelectrode and photoelectrode. These materials should be optimally placed to ensure sufficient light exposure, integrating photocatalytic materials into PBESs to increase pollutant biodegradation, further improving the removal of emerging contaminants and energy production (Li et al. 2024; He et al. 2022). Key characteristics of photocatalytic materials, such as type, high surface area, stability, and reusability, are essential for reducing the risk of secondary pollution in the environment. Nano-photocatalyst materials with magnetic properties are particularly beneficial (Zheng et al. 2024). Investigating environmentally friendly synthesis techniques for photocatalysts that utilize biological materials has gained momentum. This strategy not only minimizes dependence on toxic chemicals, but also enhances the biocompatibility of the materials, making them suitable for various applications, including wastewater treatment and energy generation (Yang et al. 2020).

Moreover, semiconductor photocatalysts can enhance the performance of PBESs by combining photoexcited electrons generated by the photocatalyst with bio-generated electrons at the cathode (Rahman et al. 2023; Rafieenia et al. 2024; Rafieenia et al. 2022). This integration allows for improved oxygen reduction, increased hydrogen evolution, and more effective pollutant removal than low-conductor PBESs (Rahman et al. 2023). For example, a solar-driven photo-bioelectrochemical cell (s-PBEC) utilizing a $\text{Co}_3(\text{PO}_4)_2/\text{Mg}(\text{OH})_2$ photocatalyst anode exhibited improved electricity generation performance (Rafieenia et al. 2024).

Rahman et al. (2023) have recently shown that photocatalysts in the cathodes of PBESs, including materials such as graphitic carbon nitride, bismuth oxy-halides, and metal oxides, are promising due to their abundance on Earth. These materials have tunable bandgaps and high efficiency, making them well suited for PBES applications. Furthermore, heterojunction and Z-scheme photocatalyst composites can be employed to improve the performance of these PBESs further (Rahman et al. 2023). Combining different photocatalyst materials in these advanced composites allows for enhanced efficiency and functionality compared to using

individual photocatalysts alone. The photocatalytic process creates electron-hole pairs that can improve the ability of microbes to transfer electrons outside the cell and onto the electrode surface, accelerating the degradation of persistent organic pollutants (Rafieenia et al. 2022).

Rahman et al. (2023) have also acknowledged that research in the field of PBESs is hindered by several key gaps, including the scarcity of studies focusing on visible light-driven photocatalysts for use in PBES cathodes and the unexplored potential of photocatalytic air cathodes for creating self-sustaining PBESs (Rahman et al. 2023). As observed in Table 1, only one study has utilized an air cathode for TCP removal in PBESs.

pH

During the operation of the PBES, organic matter near the anode is converted into protons and electrons by the oxidation process. While electrons reach the cathode from an external source, protons reach the cathode from a neutral medium and must pass through the electrode material (PEM) or salt bridge in a dual-chamber setup. Meanwhile, cathode reactions in BESs combine protons and electrons with oxygen to form water, as shown in the air cathode configuration (Zhang et al. 2019c). In biological and photoelectrochemical systems, pH is an important parameter that affects metabolic activity, growth, cellular structure, and stability of microbial biofilms on electrodes, electron production, internal resistance, elemental circulation, energy production (electricity and solar), and enzyme function (Yang et al. 2020). To achieve optimum performance, it is essential to maintain a stable pH using buffering agents or other methods. For instance, the decomposition of hard-to-degrade contaminants (e.g., carbamazepine and clorfibric acid) in a system integrating persulfate, light, and bioelectrochemical processes is efficient at neutral pH levels. In contrast, other methods, such as bio-electro-Fenton (BEF), commonly require low pH. Zou et al. (2021) reported that ultraviolet (UV) radiation used in a bioreactor provided sulfate and hydroxyl radicals, resulting in high efficiency (Zou et al. 2021). Deviations from the bacterial source pH (neutral pH effluent) during biofilm formation and function reduce the bio-electrocatalytic activity (Patil et al. 2011).

The pH value regulates the production of electrons and protons in the bioreactor, influencing the metabolic activity of microorganisms. Low pH, slightly below neutral, can inhibit metabolic activity and affect final products such as methane, as acidic pH disrupts equilibrium among composition reactions. High pH can decrease proton concentration, making it unsuitable for electron transfer (Zhang et al. 2019c). In algal PBESs, removing organic compounds and nutrients, the oxygen produced by algal activity is introduced to the cathode reaction of

Table 1 Performance overview of PBEs for ECs removal

Compounds	Bio-photoelectrode/chamber	Electrode material	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
Oxytetracycline (OTC)	Bioanode–biocathode, photocathode/two-chamber	Anode and cathode: carbon felt	Anaerobic sludge with <i>Chlorella vulgaris</i>	Visible	DC, 0.45 V	75% at 50 mg L ⁻¹ Time=160h	Photosynthesis, electrochemical reactions, mineralization	Dehydroxylated derivatives, aromatic compounds, acetic acid	(Sun et al. 2019)
Florfenicol	Bioanode–biocathode, photocathode/two-chamber	Anode and cathode: graphite	Anaerobic sludge with <i>Chlorella vulgaris</i>	Solar	DC, 100 F capacitor	93% at 5mg L ⁻¹ Time=80h	Photosynthesis, electrochemical reactions, degradation of antibiotic, nitrogen removal	Oxygen, ROS, organic compounds, nitro-gen compounds	(Li et al. 2020a, b)
Florfenicol	Bioanode–biocathode, photocathode/two-chamber	Anode and cathode: graphite	Anaerobic sludge with <i>Chlorella vulgaris</i>	LED	DC, 0.6 V	98% at 5 mg L ⁻¹ Time=95h	Substrate utilization, microalgae–bacteria interaction, electrochemical reactions	Phenolic compounds, amines, acetic acid	(Sun et al. 2020)
Metronidazole (MNZ)	Bioanode, photocathode/two-chamber	Anode: graphite felt cathode: graphite felt–Mo–W	Effluent from acetate-fed MFCs	Visible	DC, 200 Ω resistor	94.5% at 80 mg L ⁻¹ Time=1 h	Initial photodegradation, formation of intermediates, further degradation (dealkylation and ring cleavage), mineralization	Hydroxymethyl metronidazole, metronidazole aldehyde, nitroso metronidazole, carboxylic acids	(Wang et al. 2019a)
Tetracycline	Bioanode, photocathode/two-chamber	Anode: steel mesh Cathode: steel mesh–ZnIn ₂ S ₄ /Ag/AgCl/GO	Shewanella sp.	UV–visible	DC, 100 Ω	70% at 20 mg L ⁻¹ Time=4 h	Light irradiation, electron transfer, pollutant degradation, microbial action, energy recovery	–	(He et al. 2018)
Ofloxacin	Bioanode–photocathode/dual chamber	Anode: PANi@CNTs/SS Cathode: LiNbO ₃ /CF	Microbial consortium	UV	DC, 0.61 V	86.5% at 0.1 mmol/L Time=7 h	Initial photocatalytic activation, oxidation reactions	Hydroxylated compounds, carboxylic acids, amino acids	(Xu et al. 2020a, b)
Acid orange 7	Bioanode, photocathode/single-chamber	Anode: carbon brush Cathode: BiOBr	Domestic wastewater	Solar	DC, 0.6 V	90.97% at 50 mg L ⁻¹ Time=10h	Electrochemical reactions, photocatalytic activation, microbial metabolism	Smaller aromatic compounds, hydroxyl radical	(Qin et al. 2020)

Table 1 (continued)

Compounds	Bio-photoelec-trode/chamber	Electrode mate-rial	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
Methyl Orange (MO)	Bioanode, photocathode/two-chamber	Anode: graphite felt Cathode: graphite felt-Mo-W	Effluent of separate acetate-fed MFCs	Visible	DC, 0.7	100% at 20 mg L ⁻¹ Time=2 h	Anaerobic treatment, production of reactive species, oxidation reaction, photo-assisted mechanism	Benzene, hydroxyl radicals, smaller aromatic compounds	(Wang et al. 2019b)
2,4-DCP	Bioanode, photocathode/single-chamber	Anode: carbon brush Cathode: BiOBr	Domestic wastewater	Solar	DC, 0.6 V	81.41% at 20 mg L ⁻¹ Time=10h	Electrochemical reactions, photocatalytic activation, microbial metabolism	Short-chain fatty acids, carboxylic acids, alcohol and aldehydes, ketones	(Qin et al. 2020)
2,4,6- trichloro-phenol (TCP)	Photo-bioanode, air cathode/ single-chamber	Anode: mpg-C3N4-carbon felt Cathode: air cathode Cathode: Polyvinylidene fluoride (PVDF)-activated carbon	Municipal sewage	Visible	DC, 0.41 V	79.3% at 200 mg L ⁻¹ Time=10 h	Photocatalytic activation, microbial electrochemical degradation	2-Chloro-4-keto-2-hexenedioic acid (2-CMA), catechol and hydroquinone, carboxylic acids	(Wang et al. 2019a, b, c)
CAP	Bioanode, photocathode/single-chamber	Anode: carbon brush Cathode: BiOBr	Domestic wastewater	Solar	DC, 0.6 V	78.47% at 20 mg L ⁻¹ Time=10h	Photocatalytic activation, microbial degradation, electrochemical reactions	Short-chain fatty acids, carboxylic acids, alcohol and aldehydes, ketones	(Qin et al. 2020)
Congo red	Bioanode-biocathode, photocathode/two-chamber	Anode and cathode: AQDS/Mn/PPy	Anaerobic sludge with <i>Chlorella vulgaris</i>	Fluorescent	DC, 0.8 V	98% at 300 mg L ⁻¹ Time=22 h	Light absorption, electron transfer, formation of reactive intermediates	Reduced AQDS, aromatic amines, hydroxylaniline compounds, phenolic compounds	(Sun et al. 2017)



Table 1 (continued)

Compounds	Bio-photoelec-trode/chamber	Electrode mate-rial	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
Congo red	Bioanode-bio-cathode, photocathode/ two-chamber	Anode and cath-ode: graphite felt	Municipal sludge with <i>Chlorella vulgaris</i>	Fluorescent	DC,0.3 V	93% at 300 mg L ⁻¹ Time=33 h	Microbial reduc-tion, intermed-iate degradation, mineralization, bioelectricity generation	Aromatic amines, phenolic compounds, aliphatic acids (e.g., 3,4-diami-nonaphthalene-1-sulfonic acid, naphthalene-1,2-diamine, naphtha-lene-1,2,4-tricarboxylic acid, pentanal, 2-methoxy phenol	(Sun et al. 2015)
MO	Bioanode, photo-cathode/single-chamber	Anode: graphite brush Cathode: Ni foam/TiO ₂	Domestic waste-water	UV	DC,0.8 V	96% at 50 mg L ⁻¹ Time=8 h	Photocatalytic activation (e.g., light absorption and charge separa-tion), formation of reactive oxygen species (ros), hole participation, dye degradation and subsequent break-down into smaller organic com-pounds, energy production	Amines, phenolic compounds, carboxylic acids, CO ₂ , H ₂ O	(Hou et al. 2017a)
MO	Bioanode, photo-cathode/single-chamber	Anode: carbon felt/graphite, Cath-ode: Pd-modi-fied silicon	Anaerobic sludge from UASB	Visible	DC,0.4 V	84.5% at 25 mg L ⁻¹ Time=36 h	Initial photocata-lytic reaction with generating elec-tron-hole pairs, attacking hydroxyl radicals on mo-formation of inter-mediate, final mineralization	Organic by-products with molecular weights of m/z 290, 158, 123, 276	(Han et al. 2017)

Table 1 (continued)

Compounds	Bio-photoelec-trode/chamber	Electrode mate-rial	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
MO	Photoanode-bio-cathode/two-chamber	Anode: TiO2 nanotube array Cathode: Pt/C	Synthetic culture	UV-visible	DC, 0.74V	88.3% at 4.8 mg L ⁻¹ Time=240 min	Light absorption, charge separation, water oxidation, electron transfer to biocathode, reduction reactions at biocathode, pollutant degradation	Hydrogen, alcohols, aldehydes, carboxylic acids, oxygen	(Du et al. 2014)
MO	Photoanode-bio-cathode/single-chamber	Anode: TiO2/g-C3N4 nanosheet/graphene Cathode: carbon paper	Domestic waste-water	UV-visible	DC, -0.5V	90% at 100 mg L ⁻¹ Time=21 h	Biocathode reaction, photoanode reaction, nitrate reduction	Aromatic amines, hydroxylated compounds, aldehydes, nitrite, hydrogen ions, electrons	(Su et al. 2020)
MO	Bioanode-photo-cathode/single-chamber	Anode: graphite brush Cathode: g-C3N4/BiOBr	Domestic waste-water	UV-Visible	DC, 0.8V	97.8% at 50 mg L ⁻¹ Time=4 h	Photocatalytic activation (e.g., light absorption and formation of reactive species), dye degradation by ROS, intermediate formation, microbial activity, hydrogen production	Hydroxylated aromatic compounds, amino derivatives, smaller organic acids (e.g., acetic acid)	(Hou et al. 2017b)
Nitrofurazone (NFZ)	Bioanode, photocathode/single-chamber	Anode: carbon brush Cathode: g-C3N4/CdS	Domestic waste-water	Solar	DC, 0.7V	83.14% at 50 mg L ⁻¹ Time=4 h	Photocathode optimization, microbial community dynamics, formation of intermediates	Reduced nitro compounds, amino derivatives, carboxylic acids	(Hou et al. 2020b)
Ibuprofen	Bioanode, photocathode/two-chamber	Anode: PANi@CNTs/SS Cathode: CuInS2	Anaerobic sludge	Visible	DC, 0.95V	75.94% at 10 mg L ⁻¹ Time=35 h	Microbial action, oxidation reactions, formation of intermediates, complete mineralization	Ibuprofen hydroxylate, 2-(4-isobutyl phenyl) propanoic acid, aldehydes, ketones, acetic acid, propionic acid	(Xu et al. 2020a)



Table 1 (continued)

Compounds	Bio-photoelectrode/chamber	Electrode material	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
Phenol	Photo-bioanode, cathode/two-chamber	Anode: TiO ₂ /Ti Cathode: graphite plate	Body fluid analog medium	Solar	DC, 0.23V	62% at 20 mg L ⁻¹ Time=4 h	Formation of electrochemically active biofilm (microbial colonization and electron transfer), Photoelectrocatalytic activation (light absorption and charge separation), advanced oxidation reactions (OH formations (OH formation and pollutant degradation)	Aldehydes, carboxylic acids, alcohols, acetic acid, formic acid, CO ₂ , H ₂ O	(Bennani et al. 2015)
Phenol	Bioanode, photocathode/two-chamber	Anode: active carbon and mesh steel, Cathode:	Brackish sewage	Solar	DC, 0.112V	97.56% at 44 mg L ⁻¹ Time=2 days	Pollutant uptake, microbial degradation (aerobic and anaerobic degradation), electrochemical reactions	Organic acids (e.g., acetic acid), methane, CO ₂	(Verma et al. 2024)
Pyridine	Photoanode, biocathode/two-chamber	Anode: N-doped graphene/ α-Fe ₂ O ₃ —graphite felt (GF) Cathode: GF	Active sludge of domestic wastewater	Solar	DC, 0.21V	100% at 50 mg L ⁻¹ Time=96 h	Photoanode modification, photocatalytic reaction, pyridine oxidation, mineralization	Pyridine N-oxide, carboxylic acids, aldehydes	(Chen et al. 2021)
Atrazine	Bioanode, biophotocathode/two-chamber	Anode: graphite felt Cathode: indium tin oxide—polyethylene terephthalate (ITO/PET)	Anaerobic sludge with <i>Scenedesmus obliquus</i>	Solar	DC, 510 Ohm	100% at 10 mg L ⁻¹ Time=24 h	Light absorption, oxygen production, electron transfer	—	(Gonzalez Olias et al. 2019)
Ametryn	Bioanode, biophotocathode/two-chamber	Anode: carbon felt Cathode: γ-FeOOH—carbon felt	Simulated seawater	Solar	DC, 0.31V	98.7% at 2 mg L ⁻¹ Time=49 days	OH radical attack, electrochemical oxidation, microbial metabolism, photocatalytic reactions	Hydroxylated ametryne derivatives, dealkylated compounds, carboxylic acids, CO ₂ , H ₂ O	(Cai et al. 2023)

Table 1 (continued)

Compounds	Bio-photoelec-trode/chamber	Electrode mate-rial	Culture medium	Light source	Electrical voltage	Removal efficiency	Metabolic Pathway	By-product	Ref.
Thiacloprid	Bioanode, photo-biocathode/two-chamber	Anode: carbon Cathode: carbon	Wastewater treat-ment plant with <i>Chlorella</i> sp. and sludge	Solar	DC, 0.2 V	53.64% at 50 mg L ⁻¹ Time=12 days	Initial degrada-tion, formation of intermediates, microbial action	hydroxy-thiaclo-prid, chlorin-ated aromatics compounds, carboxylic acids	(Deng et al. 2022a, b)
Imidacloprid	Bioanode, photo-biocathode/two-chamber	Anode: carbon Cathode: carbon	Anaerobic sludge and wastewater with <i>Chlorella</i> sp.	Solar	DC, 0.2 V	61.66% at 50 mg L ⁻¹ Time=12 days	Initial hydrolysis, N-demethylation, dechlorination, oxidation, miner-alization	6-Chloroni-cotinic acid, 5-hydroxymi-cloprod, nico-tinic acid	(Deng et al. 2022a, b)



BES, neutralizing the pH of the algal growth culture by reducing oxygen on the BES cathode (Xiao et al. 2012). Neutralized pH can be obtained through reversible electrical current in the algal–bacterial PBES, which has been observed for complete decomposition and mineralization efficiency of ECs, such as azo dyes while producing a voltage in the bioreactor. Sun et al. (2015) observed that a reversible electric current quickly consumed the protons in the anode compartment, restoring the pH to 8.8–9.2, while the pH in the cathode decreased to 6.5–6.7 as a result of hydroxyl ion consumption and proton accumulation (Sun et al. 2015). Meanwhile, ciprofloxacin is removed at an initial pH of 6 by the UV-assisted electrochemical system due to the high generation of hypochlorous acid as an oxidant produced under irradiation. In contrast, removal and mineralization were lower at other initial pH values, i.e., 3 and 9. Therefore, pH plays a crucial role in the oxidation process of emerging contaminants (Yañez-Rios et al. 2020).

Additionally, pH affects internal resistance and electron transfer rate by changing the state of ions, such as heavy metals, in bioreactors. Low pH dissolves heavy metals, lowering internal resistance and increasing electron transfer rate, while high pH causes heavy metals to precipitate, increasing internal resistance (Yang et al. 2020). A specific pH range is necessary for microorganisms to thrive, as pH fluctuations can impede proton transfer, leading to increased internal resistance (Zhang et al. 2019c). High internal resistance within a photo-bioelectrochemical system can decrease efficiency by increasing the energy needed to drive electrochemical reactions, resulting in lower power conversion and reduced overall performance (Yi et al. 2022; Rahman et al. 2021).

The pH value influences denitrification and sulfate reduction microbial processes by affecting nitrogen and sulfur cycles in the bioreactor. Cycles subsequently direct and regulate the rate of oxidation–reduction (redox) reactions involving electron transfer between chemical species in the system. Also, the advancement of photoelectrocatalytic systems has been impacted by changes in pH, which influence the formation of intermediate species necessary for effective CO_2 reduction and various chemical transformations (Gupta et al. 2024). Several methods can stabilize pH fluctuations in the bioreactor, including adding alkaline and acidic substances to the anode and cathode, respectively, or using buffer solutions to increase the system's capacity to withstand pH fluctuations (Zhang et al. 2019c). The pH can also stimulate the activity of microbial enzymes, such as laccase, which are involved in the degradation of pharmaceuticals. Optimizing pH values can minimize fouling in membranes used in electrochemical bioreactor systems, thereby improving system performance (Moghiseh et al. 2023).

Oxygen

A photo-bioelectrochemical bioreactor integrates microbial processes and photoelectrodes to oxidize aqueous contaminants photo-electrochemically. When light strikes the surface of a photoelectrode, it triggers the release of electrons. These electrons then react with H_2O and O_2 molecules to produce reactive oxygen species or ROS (e.g., $\cdot\text{O}_2^-$, hydrogen peroxide), which are highly effective in oxidizing contaminants. Figure 2 shows the generated ROS in PBESs (Nosaka and Nosaka 2017). After oxidation, the pollutants are converted into easily degradable by-products by microorganisms in the electrically induced bioreactor, resulting from the enhanced collaboration between microorganisms and the photoelectrochemical process (Bashir et al. 2023). Additionally, returning the generated electrons to the cathode via the electroactive biofilm can help prevent photoelectron perforation.

Oxygen plays an essential role in the photocatalytic degradation of ECs, particularly during the electron transfer and oxygen reduction reaction (ORR) in PBESs. The ORR involves the reduction of oxygen molecules at the cathode, leading to the formation of hydrogen peroxide or superoxide at low concentrations (Li et al. 2020a, b). In these combined systems, photocatalysts such as sulfur-doped carbon nitride, can improve the ORR by enhancing the activation of oxygen molecules and promoting the selective production of H_2O_2 as the desired reaction product (Tong et al. 2020; Bashir et al. 2023). Oxygen influences the electron transfer process, affecting the performance and efficiency of photo-microbial fuel cells by interfering with electron transfer at the anode and increasing the potential required for this transfer. In anaerobic environments, purple bacteria can conduct photosynthetic electron transfer within their internal cell membranes; however, oxygen inhibits this process. Oxygen

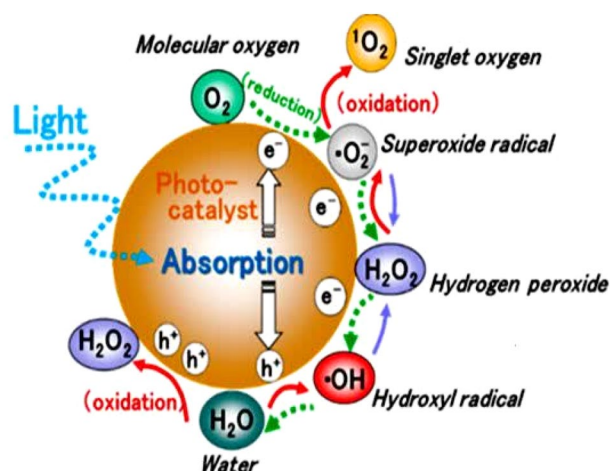


Fig. 2 Types of reactive oxygen species (ROS) produced in PBESs (Nosaka and Y. Nosaka)

functions as an electron acceptor, reducing the availability of electrons for the anode and decreasing the overall efficiency of the photomicrobial fuel cell (Voloshin et al. 2023; Huo et al. 2024).

Oxygen generation is also an essential factor in the operation of PBESs. In this bioreactor, microorganisms like algae and cyanobacteria generate oxygen through photosynthesis, transforming light energy into chemical energy, decomposing water molecules, and releasing oxygen as a by-product on the bioanode surface. This oxygen is then utilized in the cathode reaction to generate reactive oxygen species, which are crucial for the system's overall performance and efficiency (He 2015; Xiao et al. 2012). Research has focused on photovoltaic systems, emphasizing their ability to utilize oxygenic photosynthetic organisms—like microalgae and cyanobacteria—to convert light energy into electrical power without organic substrates. These systems rely on the natural photosynthetic processes that generate oxygen as a by-product, which is then utilized in electrochemical reactions. This approach contributes to energy harvesting and environmental sustainability by reducing reliance on organic fuels (McCormick et al. 2015). However, oxygen concentration should be optimized for the degradation and mineralization of ECs. Zhu et al. (2021) have acknowledged the critical influence of N-TiO₂ photocatalyst application on decreasing oxygen diffusion in the air cathode, enhancing bioanode performance and radical production in a BES–photocatalysis reactor. This photocatalyst achieved the appropriate concentration of oxygen for decomposing 2,4,6-trichlorophenol (TCP) (Bashir et al. 2023; Zhu et al. 2021).

Electrochemical water electrolysis offers a promising method for generating oxygen, which can support cell functions in the bioreactor. Lee et al. (2023) used an electrocatalytic method for oxygen generation in the therapy area of the cell (Lee et al. 2023). The oxygen generated through photosynthesis in PBESs has a complex impact on the system. Additionally, oxygen is generated through the photocatalytic activity of thylakoid membranes extracted from spinach in PBESs. These membranes are mixed with a ferricyanide buffer solution and introduced into the cell. Upon simulated sunlight exposure, water oxidation and transfer of electrons from the thylakoids to a transparent electrode occur. This process generates a photocurrent density of 0.5 mA/cm² (Pinhassi et al. 2016).

Applied voltage

The role of voltage in removing emerging contaminants in PBESs is crucial for effective wastewater treatment and environmental remediation. Applying an external voltage can enhance the photocatalytic efficiency of electrode materials in BESs. The research has shown that introducing a small external voltage of around 0.5 V can decrease

the recombination rate of electron–hole pairs generated by light on the surface of the photocatalyst as demonstrated in Fig. 1, leading to improved degradation efficiency (~80% removal) of pollutants like chlorophenols, surpassing both the unirradiated BES (66.0%) and the photocatalytic-only process (56.1%) (Tong et al. 2022; Zhu et al. 2021). This study demonstrates that the synergy between photocatalysis and BESs enhances contaminant removal efficiency and boosts electricity production by transferring electrons from the microbial biofilm to the photocatalyst. The voltage plays a significant role in the performance of both photocatalysis and BESs, with higher voltages increasing the cathode's potential and improving pollutant degradation. Extremely high voltages can reduce cathode performance due to energy losses and potential damage to microbial communities involved in the bioprocess. Zhang et al. (2022) noted that excessive voltages could inhibit system performance, highlighting the delicate balance required when selecting operational voltages for optimal pollutant degradation (Zhang et al. 2022). Table 1 identifies optimal voltage ranges (typically between 0.2 and 0.8 V) that balance microbial activity and electrochemical efficiency. Cai et al. (2023) obtained a significant removal efficiency of 98.7% for ametryn over 49 days, which is six times higher than the natural degradation rates. This enhancement is attributed to the continuous generation of oxidative species facilitated by the steady voltage of 0.31 V during operation (Cai et al. 2023). The study shows that a voltage of 0.2 V enhances microbial activity and improves electron transfer processes. As a result, this leads to a more effective breakdown of the thiacloprid pesticide. Consequently, the degradation process becomes more robust, allowing for higher removal efficiencies of thiacloprid than systems operating at lower voltages (Deng et al. 2022a).

In BESs, the voltage is influenced by light conditions. The optimal voltage for photocatalysis is usually attained under specific lighting conditions, which depend on the type of photocatalytic material used and its photoelectrocatalytic activity. This underscores the importance of considering both voltage and power density, as well as light conditions (particularly UV–visible light), for efficient pollutant degradation (Tong et al. 2022). For example, a study using BaTiO₃ as the photocathode found that the open circuit voltage (OCV) increased from 280 mV to 387 mV when exposed to UV–visible light. Additionally, the maximum power density rose from 64 mW/m² to 498 mW/m², representing a significant 7.8-fold increase compared to the value measured without light (Touach et al. 2023). In another study using TiO₂ as the photocathode, the OCV improved from 0.0465 V to 0.3979 V, and the power density increased from 54.88 mW/m² to 70.39 mW/m² after modifying the TiO₂ in the photocathode (Tong et al. 2022).

Light source

Besides the photocatalytic material used in the bioreactor, under light conditions, the internal resistance of the device was 443.4Ω , significantly lower than the 1378Ω reported in the absence of light. This reduction in internal resistance enhances the power density and efficiency of the cell, ultimately improving pollutant degradation (Tong et al. 2022).

Another important factor is light conditions such as intensity and wavelength. The wavelength of light can affect the efficiency of pollutant degradation. For example, visible light can be utilized for photocatalytic reactions, as demonstrated in a study by Yan et al. (2018), which used a membraneless photocatalytic fuel cell with redox reactions of H_2O_2 to detect a type of tetrachlorobiphenyl using visible light (Yan et al. 2018). However, UV light significantly enhances the efficiency of decolorization and hydrogen production compared to conditions without UV light. This improvement is attributed to the higher energy of UV light, which leads to the generation of more reactive species commonly utilized in photocatalytic reactions. Nonetheless, UV light can also be more toxic to microorganisms, necessitating a balance between photocatalytic efficiency and microbial viability (Tong et al. 2022; Yi et al. 2022). Although the study found that increased light intensity improved the efficiency of pollutant degradation, it should not be so high as to harm the growth and activity of the electroactive microbial biofilm in PBES. It should be uniformly distributed across the photocatalytic cathode surface (Voloshin et al. 2023). The ideal light intensity for photocatalytic microbial electrolysis cells varies depending on the specific system configuration and the type of photocatalytic material employed (Yi et al. 2022; Bell et al. 2013). Table 1 shows that solar lighting has the highest consumption compared to LED and fluorescent lighting, according to various studies. The findings highlight the importance of light as an energy source for microalgae, significantly impacting both their metabolic activity and the efficiency of pesticide degradation in BESs (Deng et al. 2022a, b), while Sun et al. (2017) used fluorescence to stimulate the photo-bioelectrochemical processes within the cell through photogenerated electron transfer processes for enhancing both bioelectricity generation and the degradation of Congo red (Sun et al. 2017). Although working conditions such as applied voltage, electrodes, and contaminants were different in both studies, the microbial culture was applied an algal-bacterial biofilm.

Temperature

Carefully considering the influence of temperature, in addition to other key factors, is crucial for optimizing the overall efficiency and performance of both the photocatalytic and microbial components within PBESs. Temperature

significantly influences the light response of photocatalytic materials. For instance, a study demonstrated the importance of temperature in enhancing the light response. They used a photocatalytic two-chamber BES with a rutile-coated graphite photocathode. They observed that the output voltage rose from 337 to 384 mV when the system was exposed to simulated natural light conditions. This underscores the significant impact of temperature on the light response of photocatalytic materials (Omidi and Sathasivan 2013; Chen and Hsu 2021; Tong et al. 2022). The kinetics of electrochemical reactions are influenced by temperature. Higher temperatures can result in faster reaction rates, potentially improving current generation in PBES (Yang et al. 2020).

Additionally, temperature can enhance enzymatic and microbial activity and growth, as well as provide the kinetic energy necessary for the effective biodegradation of organic contaminants. This leads to increased removal rates, particularly at higher temperatures. However, very high temperatures can negatively affect microbial viability and removal rates. It is essential to maintain an optimal temperature range, typically between 35°C and 40°C , to ensure the efficient performance of the microbial component in PBES (Omidi and Sathasivan 2013; Chen and Hsu 2021; Tong et al. 2022).

Furthermore, the optimal temperature can improve the charge carrier mobility and interfacial charge transfer between the photoelectrode and microorganisms, enhancing the removal rate (Chen and Hsu 2021). A study using an $\text{LiTaO}_3/\text{g-C}_3\text{N}_4$ heterojunction as a photocathode in a PBES demonstrated the importance of temperature in enhancing pollutant removal rate and system performance. The study found that when the system was exposed to light at a higher temperature (30°C), the removal efficiency of chemical oxygen demand (COD) significantly increased compared to lower temperatures (20°C) (Lazar Nour-eddine 2024). The optimal temperature ranges for achieving optimal contaminant removal can vary depending on the PBES, the microbial species, and their components in the bioreactor. For instance, a study using a BaTiO_3 photocathode found that the optimal temperature for pollutant removal was around $30\text{--}35^\circ\text{C}$. At this temperature, maximum removal efficiencies for contaminants were achieved. One study reported a COD removal efficiency of 93.5% at 35°C , notably higher than that at lower or higher temperatures. Generally, the performance of BESs tends to decline at temperatures below 25°C and above 40°C . Since COD removal efficiencies decreased to approximately 57.5% and 62.3% at 15°C and 20°C , respectively, there was a decline in performance due to reduced microbial activity at temperatures exceeding 40°C (Touach et al. 2023; Tee et al. 2017; Malik et al. 2022; Behera et al. 2011). Another study showed that using a TiO_2 nanotube array photoanode in PBES improved hydrogen production at 25°C and 35°C (Kim et al. 2018). For optimal performance,

it is generally understood that BESs operate effectively at ambient temperatures, typically around 20–30°C, but specific studies may vary based on the materials and configurations used. Besides, studies have shown that BESs can operate at various temperatures, including 25 ± 4 °C and 32 ± 4 °C, which are typical for investigations involving organic substrates such as leachate (Sonawane et al. 2022).

The aforementioned issues emphasize the importance of determining the appropriate temperature conditions for each PBES to maximize the pollutant degradation efficiency (Touach et al. 2023). Temperature also indirectly affects BESs by influencing pH levels, which are critical factors for microbial activity and electron transfer. The optimal pH level for microbial activity, growth, and electron transfer was around 6.88 at the environmental temperature, resulting in enhanced current production (Ishaq et al. 2024).

Conductivity

The ionic conductivity of the electrolyte in a PBES directly affects the internal resistance of the system. Higher electrolyte conductivity can decrease internal resistance, enhancing the power output and system efficiency (Murugaiyan et al. 2022; Yang et al. 2021). The type, concentration, and chemical composition of the electrolyte solution are essential factors that influence the operation of a bioreactor. Researchers have shown that employing saline electrolytes, specifically sodium sulfate (Na_2SO_4) solutions, can enhance ionic conductivity and decrease internal resistance in PBESs (Yang et al. 2021; Tong et al. 2022). Potassium phosphate buffer solutions have also been used as electrolytes in PBES configurations, offering high ionic conductivity and a suitable pH environment that effectively support microbial and photocatalytic processes (Tong et al. 2022). On the other hand, Najafgholi et al. (2015) modified electrical conductivity by adding NaCl and discovered that optimal conductivity levels significantly improved the power density and current output. They achieved a peak power density of 10.08 mW/m² at a conductivity of 1.053 mS/cm, which increased with higher concentrations of NaCl (Najafgholi et al. 2015). In a study that used pineapple waste in single-chamber bioelectrochemical systems (BESs), researchers found a maximum power density of 513.99 ± 6.54 mW/m² at an optimal substrate electrical conductivity of 145.16 ± 9.86 mS/cm. This research emphasizes the critical role of substrate conductivity in improving bioelectricity production from organic waste (Rojas-Flores et al. 2022). In the second study, despite the large amount of substrate, it produced a significant amount of current density. A study indicated that raising the solution conductivity to 20 mS/cm significantly improved hydrogen production rates when operating at applied voltages between 0.3 V and 0.6 V (Wu et al. 2017).

Additionally, research on BESs revealed that decreasing the pH of the electrolyte can increase its conductivity, highlighting the potential for pH adjustments to enhance ionic flow (Yang et al. 2021). However, the overall effect of pH adjustments on PBES performance requires detailed assessment to ensure optimal operation.

Meanwhile, electrical conductivity (EC) is a key indicator to assess contamination levels in aqueous solutions. In PBESs, EC is crucial for monitoring the system's efficiency in removing contaminants. Higher EC levels indicate higher contamination amounts, which can be mitigated by optimizing the system's parameters to improve the performance. Research has demonstrated a direct relationship between EC and the concentrations of specific emerging contaminants, including pharmaceuticals. This correlation suggests that EC can be used to monitor regions with high contamination levels, enabling more comprehensive analysis and remediation efforts.

Studies have shown that optimum EC levels can affect the rate of photocatalytic reactions at the cathode and redox reactions in PBESs by facilitating the mobility or transfer of ions and electrons and reducing ohmic losses between the electrodes and microbial cells. Conductivity also affects microbial activity, growth, and metabolic processes related to emerging contaminant bioavailability, subsequently impacting the removal efficiency of contaminants. Carbon felt is often preferred due to its high porosity, which promotes microbial attachment and biofilm formation (de Sousa et al. 2014; Rousseau et al. 2020; Radhika et al. 2022; Rafieenia et al. 2022; Wu et al. 2017).

The study results indicate that increasing electrical conductivity, such as by adding seawater, can improve the removal of other pollutants like COD, total suspended solids (TSS), turbidity, and color in an electrocoagulation (EC) treatment process. However, the addition of seawater, which increases electrical conductivity, also led to higher levels of pH, dissolved oxygen (DO), and electrical conductivity itself (Ahmed et al. 2024). If conductivity is very high, it can prevent the critical issue of anode acidification in PBESs caused by the oxidation of organic matter by electroactive bacteria.

Therefore, there should be a balance between the electrolyte solution and other conditions, especially optimal pH for the bioanodes in PBESs. Researchers have reported that PBESs perform effectively using an electrolyte with a conductivity of $10.7 \text{ S} \cdot \text{m}^{-1}$ (Rousseau et al. 2020). The current density observed in PBESs serves as a direct indicator of their conductivity performance. Studies show that current densities can vary between 0.15 A/m^2 and 0.38 A/m^2 under ideal conditions, highlighting the potential for significant electricity generation by integrating photocatalysis with electroactive microbes (Rafieenia et al. 2024).

Substrate

Emerging contaminants such as metronidazole, ciprofloxacin, ethinylestradiol, p-nitrophenols, oxyfluorfen, fipronil, 4-chlorophenol, methyl orange, and ethanolamine can be used as substrates in BES to generate the final products such as hydrogen gas, electricity, and valuable materials, effectively eliminating these contaminants. This integrated photocatalysis and BES process enhances the effect compared to using photo-catalysis, microbial fuel, or electrolysis cells alone. This process also reduces the toxicity of these compounds to electroactive microorganisms, which results from their low biodegradability (Thulluru et al. 2023). A summary of the literature review on the performance of PBESs with different reactor schematics, including the removal of ECs as substrates, is presented in Table 1.

As shown in Table 1, a study utilized PBES with a bioanode and a photocatalytic cathode to remove ibuprofen (IBU) while simultaneously generating power. Under optimal conditions, researchers achieved 75.94% removal and 65.1% mineralization efficiency for IBU at a concentration of 10 mg L⁻¹, with a maximum power density of 0.119 W m⁻², a current density of 0.75 A m⁻², and a voltage of 950 mV. They found that the most reactive oxidative species were O₂ and OH⁻, indicating the electron transport mechanism and the ibuprofen (IBU) decomposition pathway in the system, as illustrated in the following section (Xu et al. 2020a).

Studies have reported that emerging contaminants can significantly impact the performance and efficiency of bioreactors due to their complex structures, which are more challenging to biodegrade compared to more straightforward organic matter such as acetate. These complex structures can stimulate microbial diversity in the anode biofilm and electron transport chains (Amar Dubrovin et al. 2022; Hua et al. 2019). Generally, organic material (e.g., pharmaceutical wastewater) acts as an electron donor in electrochemical reactions, oxidized by electroactive microbial biofilm to generate electrons, which are then used to reduce protons at the cathode, resulting in the production of hydrogen gas and electricity in PBESs (Tong et al. 2022; Jeon et al. 2018; Parvanova-Mancheva et al. 2022). These findings indicate that the optimum concentration or organic load of emerging contaminants is crucial for the bioreactor's performance, as excessively high concentrations can be toxic and inhibitory to the metabolic and enzymatic processes of the microbial community (Yi et al. 2022).

The search results do not often address the application of PBESs and algae-based photosynthetic microbial electrolysis cells in the removal of pharmaceutical and hormone compounds. However, the results emphasize the potential of integrated biological and electrochemical treatment methods, such as the high-rate algal pond (HRAP) coupled with electro-oxidation (EO), to achieve high removal rates for

various pharmaceutical substances. Additional research is needed to evaluate the effectiveness of photo-assisted microbial electrolysis cells in this specific application. A combined biological and electrochemical treatment approach using a high-rate algal pond (HRAP) followed by an electro-oxidation (EO) process with a boron-doped diamond (BDD) electrode achieved removal rates exceeding 80% for drugs used to lower lipid levels (hypolipidemic), relieve pain (analgesic), and reduce inflammation (anti-inflammatory). Conversely, antiepileptic drugs such as carbamazepine exhibited lower removal rates, ranging from 36.8–69.1% (Jiménez-Bambague et al. 2023).

Algae-based PBESs allow for secondary and tertiary domestic wastewater treatment while generating bioelectricity. Microalgae and bacteria in this method have a symbiotic relationship that can effectively remove nitrogen, phosphorus, and organic matter from the wastewater (Mansoor et al. 2023). Algae-based PBESs have shown the capability to effectively eliminate organic pollutants at high rates, up to 381.43 gr/m³.d, and have also successfully removed nutrients such as nitrogen and phosphate (Yi et al. 2022).

Another emerging contaminant is pesticides. Studies have reported over 95% removal of various pesticides, including organophosphates, pyrethroids, and organochlorines, using algae-based systems. These cost-effective and environmentally friendly approaches can remove contaminants through processes like biosorption to the algal biomass, bioaccumulation within the algal cells, and biodegradation of complex organic compounds into simpler ones (Riaz et al. 2017; Abdelfattah et al. 2023). According to Table 1, algae-PBESs can efficiently remove pesticides from water, even at very low concentrations (as small as 0.1 µg L⁻¹), sufficient to comply with regulatory standards for pesticide levels in water (Gonzalez Olias et al. 2019). The efficiency of removing pesticides using a combination of solar photo-Fenton and biological treatment depends on the initial pesticide concentration. When the initial concentration was higher (C₀=500 mg/L), a longer solar photo-Fenton treatment time is required to achieve sufficient mineralization (55%) for effective biodegradation compared to lower concentrations (C₀=200 mg/L, 33% mineralization) (Martín et al. 2009). According to Table 1, Cai et al. (2023) and Deng et al. (2022a) have shown that ametryn (as the herbicide) and thiacloprid (as an insecticide) have high efficiency at concentrations of C₀- 2 mg L⁻¹ and C₀- 50 mg L⁻¹ (Cai et al. 2023; Deng et al. 2022a). These findings indicate that higher pesticide loads can decrease the biodegradability of the intermediate compounds generated during the photocatalytic treatment process (Martín et al. 2009). Neonicotinoid insecticides-such as imidacloprid, clothianidin, acetamiprid, and thiamethoxam-were mineralized up to 80%, with degradation reaching as high as 96% through enhanced

removal methods combining electrochemical oxidation and biodegradation. The integrated technology generates low-toxicity intermediates during the electrochemical process, which provided carbon sources for bacterial growth to strengthen the mineralization of the investigated contaminants (Satheeshkumar et al. 2024).

However, the search results do not provide information on the use of photocatalytic microbial electrolysis cells for pesticide removal, nor do they specify how pesticide concentration affects the efficiency of this treatment approach. In addition to the above-mentioned compounds, phenolic compounds and dyes are also discussed in Table 1. As shown in Table 1, PBESs can completely degrade phenolic compounds such as 2,4,6-trichlorophenol (TCP) within 72 h. The degradation rates achieved by PBESs are 1.8 times higher compared to microbial fuel cells that do not utilize photocatalysis (Tong et al. 2022). Aniline and phenol were also removed by photoelectrocatalytic BES (PEC-BES) degradation, with high removal rates reported for this process at 70% and 96% for 165 and 700 mg L⁻¹, respectively (Zhang et al. 2019b). Moreover, dyes can serve as alternative electron sources to drive electrochemical reactions that generate hydrogen gas in these bioelectrochemical systems, as frequently shown in Table 1. Acid orange 7 (AO7), 2,4-dichlorophenol (2,4-DCP), and chloramphenicol (CAP) as industrial additives were substrates of single-chamber BPES with a BiOBr photocathode and anodic biofilm, achieving high TOC removal rates of 90.97%, 81.41%, and 78.47% at an initial concentration of 50 mg L⁻¹, respectively (Qin et al. 2020).

Similarly, Hou et al. (2017b) concluded that methyl orange (MO) decolorization efficiency reached 96% at an initial concentration of 50 mg L⁻¹ during an 8-h retention time in a single-chamber PBES. Photocatalytic materials can accelerate azo dye degradation and concurrent hydrogen production by enhancing electrochemical reactions in the bioreactor. However, the efficiency decreased to 76% with an increasing concentration of 300 mg L⁻¹ (Hou et al. 2017b). Other studies also reported removing MO from wastewater, as represented in Table 1.

Sun et al. (2015) reported 100% decolorization and 50% mineralization rates of congo red (with an initial concentration of 300 mg L⁻¹) in 22 h in algal-bacteria PBESs, which was an 8% increase in mineralization rate compared to algae-free BESs. This system used a mixture of algae and bacteria to grow as a biofilm on both the anode and cathode electrodes. The algae species used was *Chlorella vulgaris*, and the bacteria came from indigenous microorganisms naturally present in the wastewater used as the inoculum or starter culture (Sun et al. 2015). Photosynthetic organisms like algae or photosynthetic bacteria can provide additional reducing power and electron donors to

drive the BES reactions for dye removal and hydrogen production, as shown in Table 1 (Yi et al. 2022).

Predominant microorganisms

Electroactive microorganism communities on the electrodes function as anode or cathode in PBESs, which play a crucial role in enhancing the degradation and treatment of organic contaminants while simultaneously producing energy, electricity, or chemical matter through their metabolic activities. Additionally, electroactive microorganisms provide mixed microbial cultures that can contribute to the electron transformation chain. Furthermore, biological growth and reactions involving photochemical processes in PBESs are recognized as essential factors in the degradation of ECs. Yi et al. (2022) have reviewed these microorganisms in PBESs, including *Geobacter* and *Shewanella* species. It appears that they used four routes for the electron transformation chain: cytochromes, nanowires, oxidation–reduction reactions, and metabolic by-products. Additionally, they reported that photosynthetic bacteria (e.g., *Rhodobacter*) and microalgae (e.g., *Chlorella*) can generate oxygen and organic matter through the process of photosynthesis, which contributes to the growth of electroactive microorganisms and energy production. Methanogen (i.e., *Methanosaeta* and *Methanococcus*) and hydrolytic microorganisms (i.e., *Clostridium*), which help in converting organic matter into methane as a biogas and energy source, and nitrifying (*Nitrosomonas* and *Nitrobacter*) and phosphate-accumulating bacteria, which are crucial in recycling nitrogen and phosphorus from wastewater (Yi et al. 2022). Similarly, in other studies, *Geobacter* and *Shewanella* species are the electroactive microorganisms in BESs, which degrade ECs (i.e., phenol) because they provide pili or conductive nanowires adhering to electrodes. Moreover, *Pseudomonas* and *Bacillus* species are predominant microorganisms in degrading ECs such as phenol and pesticides, which can be due to the enzymatic activities of the cited microorganisms in BESs. It is worth mentioning that the performance of these microorganisms can be influenced by parameters such as substrate degradability, operational conditions, and bioreactor designs (Ahmadi et al. 2023). However, it has been reported that *Pseudomonas* and *Bacillus* can obtain benzoquinone intermediates through a mono-oxygenation process from phenolic compounds (Hasan et al. 2019).

A study by Xiao et al. has identified the predominant microorganisms involved, including microalgae (e.g., *Chlorella vulgaris*), which assist in the cathodic reactions in BESs. Also, the study mentions that bacteria cultures widely have aquatic and soil environments commonly that degrade the nutrients and organic carbon in photo-BESs (Xiao et al. 2012). Moreover, low-toxicity microbial cultures (e.g., those found in domestic wastewater) exhibit greater

diversity than high-toxicity cultures (e.g., those found in industrial sewage) in the bioreactor. The microorganisms in these cultures include aerobic and facultative anaerobes (e.g., *Pseudomonas* and *Bacillus* species), anaerobic bacteria (e.g., *Clostridium* and *Bacteroides*), and electroactive bacteria (e.g., *Geobacter* and *Shewanella*). These microbes contribute to fermentation and the breakdown of complex ring structures into simpler compounds, such as emerging contaminants (ECs). Additionally, electroactive bacteria can transfer electrons to electrodes and degrade aromatic compounds, including ECs (Hassan et al. 2019).

Reactor designs

As cited before, PBESs are integrated biological and light-driven electrochemical processes that enable different chemical transformations that generate electrons and protons at the anode. These electrons and protons then combine at the cathode to reduce electron acceptors. The design of the reactors within these systems is crucial for their application, efficiency, and overall performance affecting the microbial metabolic processes that enhance the degradation of ECs, microbial electricity, hydrogen, and other suitable chemical production. Several PBES designs—such as photoanode-biocathode, bioanode-photocathode, photo-bioanode, and bioanodephotobiocathode—have recently been developed by combining semiconductive electrodes, light sources, and photosynthetic organisms or algae (Yi et al. 2022). The researchers pointed out that the bioanode–photocathode configuration enhances the production of hydrogen or multicarbon compounds by efficiently using the photoelectric potential generated by the photocathode. Also, the photoanode removes the recalcitrant contaminants, while the biocathode performs the biological treatment processes, such as denitrification in the photoanode–biocathode design. This design boosts the efficiency of nitrate and pollutant removal; however, the consumption of the chemical material or electrical power has been eliminated in the bioreactor. The study of Yi et al. (2022) focuses on both configurations (i.e., bioanode–photocathode and photoanode–biocathode). These systems promote the purification of wastewater and the generation of bioenergy by integrating microbial action at the anode with light-driven processes at the photoelectrodes (Yi et al. 2022). According to Table 1, many studies have utilized the bioanode–photocathode configuration for removing contaminants in PBESs. For instance, Wang et al. (2019a, b, c) confirmed that this configuration involves high efficiency in the biodegradation and mineralization of metronidazole during the low time and the high concentration (Wang et al. 2019a, b, c). In addition, reactor configurations or designs in PBESs can be classified into two types: two chamber and single chamber, depending on the bioreactor's purpose or application. The dual-chamber or H-cell configuration has

often been utilized in bioreactors, as indicated in Table 1. The anode and cathode are separated by a selectively permeable membrane. This configuration could have high application and function to enhance hydrogen production. However, the anode and cathode are placed within the same chamber without membrane in the single chamber, allowing for more straightforward construction and operation. This configuration can benefit specific microbial processes where direct contact between the electrodes and microbes is useful (You et al. 2023). Rusyn et al. (2024) found the integration of constructed wetlands and BESs (CWBESs) can remove 87% and 90% of pharmaceuticals and dyes, respectively, in the two-chamber configuration. Among pharmaceuticals, this study has identified removal efficiencies of 82–100% for antibiotics and 8–96% for anti-inflammatories, accompanied by green energy production from wastewater through the evapotranspiration and radial oxygen loss mechanisms of plants and bacteria in CWBESs (Rusyn et al. 2024).

The choice of electrode materials significantly impacts the efficiency of electron transfer and microbial attachment. Conductive materials, including biochar, carbon-based materials, and conductive polymers, are commonly used to enhance performance. You et al. (2023) acknowledged a removal efficiency of 91.7% in CWBESs, the solar-assisted BES, operating in continuous mode with stainless steel mesh and carbon fiber felt as the electrode materials (You et al. 2023). Additionally, these electrodes have been used to enhance conductivity and microbial attachment, where ECs such as phenol and 3,5-bis (1,1-dimethyl ethyl) were removed by increasing bioaccumulation in plant shoots and microbial activity in multi-stage CW–BESs with halophyte plants, providing suitable conditions for plant growth and a diverse bacterial population compared to conventional CWs (Verma et al. 2024).

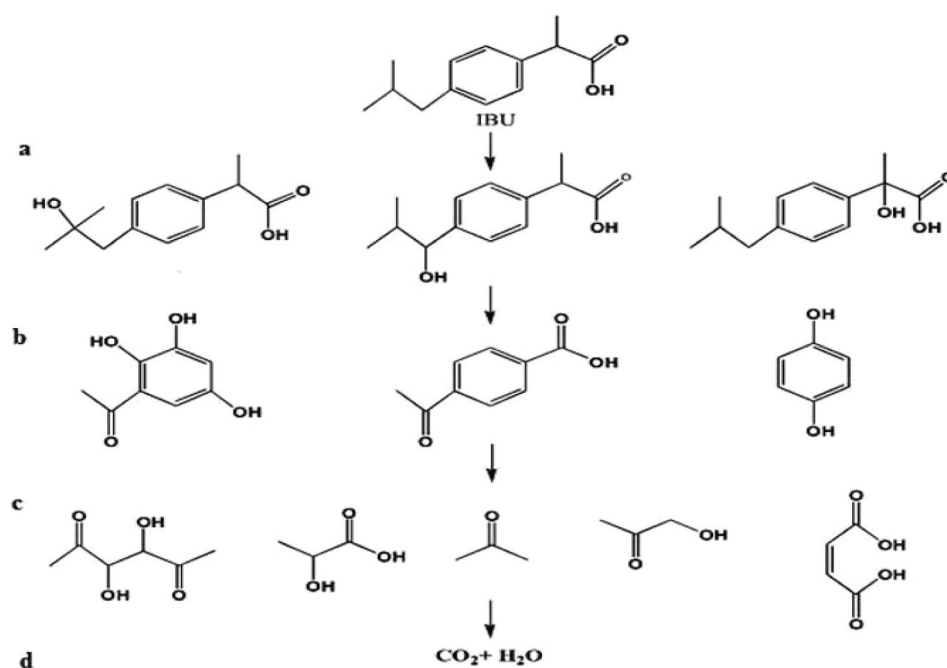
Influence of PBESs on the degradation pathway and mechanism

The presence of ECs, including intermediate or by-product residues from various sources, in the environment raises concerns about their hazardous and accumulative impacts on human health and other living organisms through the food chain, disrupting ecosystems. These impacts have been frequently observed in different studies (Moghiseh et al. 2023). ECs, such as pharmaceuticals, can generate resistant and toxic intermediate and by-product compounds that persist in wastewater due to the inefficient performance of WWTPs in removing these substances. As a result, they can cause secondary pollution in contaminated environments such as rivers, lakes, and other water sources. Therefore, the treatment processes employed are crucial, as they impact the production of intermediates and the pathways involved. PBESs can efficiently remove ECs through photocatalysis

and bioelectrochemical processes. Photocatalysis uses light energy to drive chemical reactions that degrade contaminants, while bioelectrochemical processes utilize the metabolic activities of microorganisms to decompose pollutants further. Photocatalytic degradation, performed by photocatalysts (e.g., TiO_2), generates reactive oxygen species such as hydroxyl and superoxide radicals ($\cdot\text{OH}$ and $\text{O}_2\cdot^-$) or hydrogen peroxide. These components decompose contaminants such as dyes, pharmaceuticals, hormones, and pesticides. Subsequently, electroactive microorganisms in BESs break down and oxidize organic compounds, producing electrons (Bashir et al. 2023). The electroactive microorganisms on biofilm may receive or transfer electrons from the electrode surface through electron shuttles (ESs) such as cytochromes and redox-active proteins containing an iron atom that attach to plasma and cellular membranes. Other mechanisms include conductive proteins in the exopolysaccharides (EPS), electrically conductive nanowires, oxidation–reduction agents, and metabolic by-products. It is important to mention that the electron transfer pathways at the cathode are similar to those found at the anode (Yi et al. 2022). These electrons are then transferred to the anode, generating an electric current. The induced current can stimulate and enhance the enzymatic and metabolic activity of microorganisms, acting as a biocatalyst, which increases their ability to transfer electrons and further break down pollutants, leading to improved contaminant removal. Furthermore, BESs can degrade any remaining by-products or intermediate compounds from the initial oxidation, allowing for a more thorough breakdown of the original contaminants into simpler, less harmful substances—a process known as

mineralization (Bashir et al. 2023). The impact of PBESs on the probable pathway of ibuprofen (IBU) degradation is illustrated in Fig. 3. According to Table 1, IBU is degraded by reactions between the bioanode and photocathode in the two-chamber reactor studied by Xu et al. (2020a, b). As shown in Fig. 3, they found (a) hydroxylated IBU through the interaction between $\cdot\text{OH}$ and $\cdot\text{O}_2^-$ radicals and the structure of IBU. Subsequently, simpler intermediates with low molecular weights, such as (b) decarboxylated and demethylated phenolic rings, were identified through the interaction between $\cdot\text{OH}$ and $\cdot\text{O}_2^-$ radicals and dehydroxylated IBU. These interactions continued, (c) forming short and simpler chains that were then (d) converted into water and carbon dioxide molecules (Xu et al. 2020a). Similarly, hydroxy-ibuprofen has been recognized as a significant by-product of ibuprofen breaking down into inorganic substances due to the metabolic and enzymatic activities of microorganisms in the bioreactor. Subsequently, this by-product can be further degraded into mineralized products using a combination of a BES and a constructed wetland bioreactor system (CW–BES) (Li et al. 2019). Generally, the final intermediates from decomposing the phenolic rings are reported to have lower toxicity levels compared to the primary compound, observed as inorganic and organic acids, e.g., propionic acid, oxalic acid, and acetic acid, from the biodegradation of the phenol ring using BESs supplied by direct electrical power in the study by Moghiseh et al. (Moghiseh et al. 2019). The proposed degradation pathway can be complex depending on the type of compound degraded in PBESs. For instance, Hou et al. (2020a, b) investigated the mechanism and pathway of nitrofurazone (NFZ)

Fig. 3 Possible pathways for ibuprofen (IBU) degradation (Xu et al. 2020a)



degradation, an antibiotic, in a PBES without electrical input, known as a self-biased bio-photoelectrochemical system (SB-BPES). This study identified more complicated, abundant, and toxic intermediates in the proposed degradation pathways. Figure 4 indicates the main intermediates of NFZ degradation, including six intermediates with the furan ring and two linear chains, eventually mineralized into H_2O and CO_2 . As shown in Fig. 4, pathways a, b, and c demonstrate the reduction of the nitro group of NFZ to the amino group (NH_2^-) (c), the hydroxyamino group (HOHN^-) (b), or NFZ broken down to (5-nitro-2-furyl) methanamine (NFF) (a). The intermediates from pathways a, b, and c are further reduced or broken down to methanamine. The furan ring is oxidized and then reduced, and eventually converted into 5-hydroxy-cadaverine H_2O and CO_2 . Microbial processes, including adsorption and degradation by microbes on both the anode and cathode, as well as photocatalytic degradation, contribute to the breakdown of NFZ. It can be eliminated on both the anode and cathode of a PBES reactor (Hou et al.

2020a). Thus, the combination of electrochemical and photochemical energy sources, along with the catalytic activity of microbes, enables PBESs to effectively remove contaminants from aqueous solutions and produce biofuels or electricity (Yi et al. 2022). This mechanism has occurred for other ECs in PBESs, which are depicted in different schematics. Pesticides such as ametryn, imidacloprid, and thiacloprid have been studied by Deng et al. (2022a, b) and Cai et al. (2023) in PBES. In their research, several degradation pathways have been proposed based on the intermediates produced in the system. Thiacloprid contains pyridine and thiazole rings that have been converted to seven degradation intermediates through ring breaking, hydroxyl addition, and oxidation in BESs that accompany microalgae as a biocathode, which are shown in Fig. 5. The major approach reported to reduce toxicity involves the oxidation of the thiazole ring. The intermediates in the probable pathways are shown as mass spectra in Fig. 5. Microalgal photosynthesis occurs in the biocathode chamber, producing oxygen as a by-product.

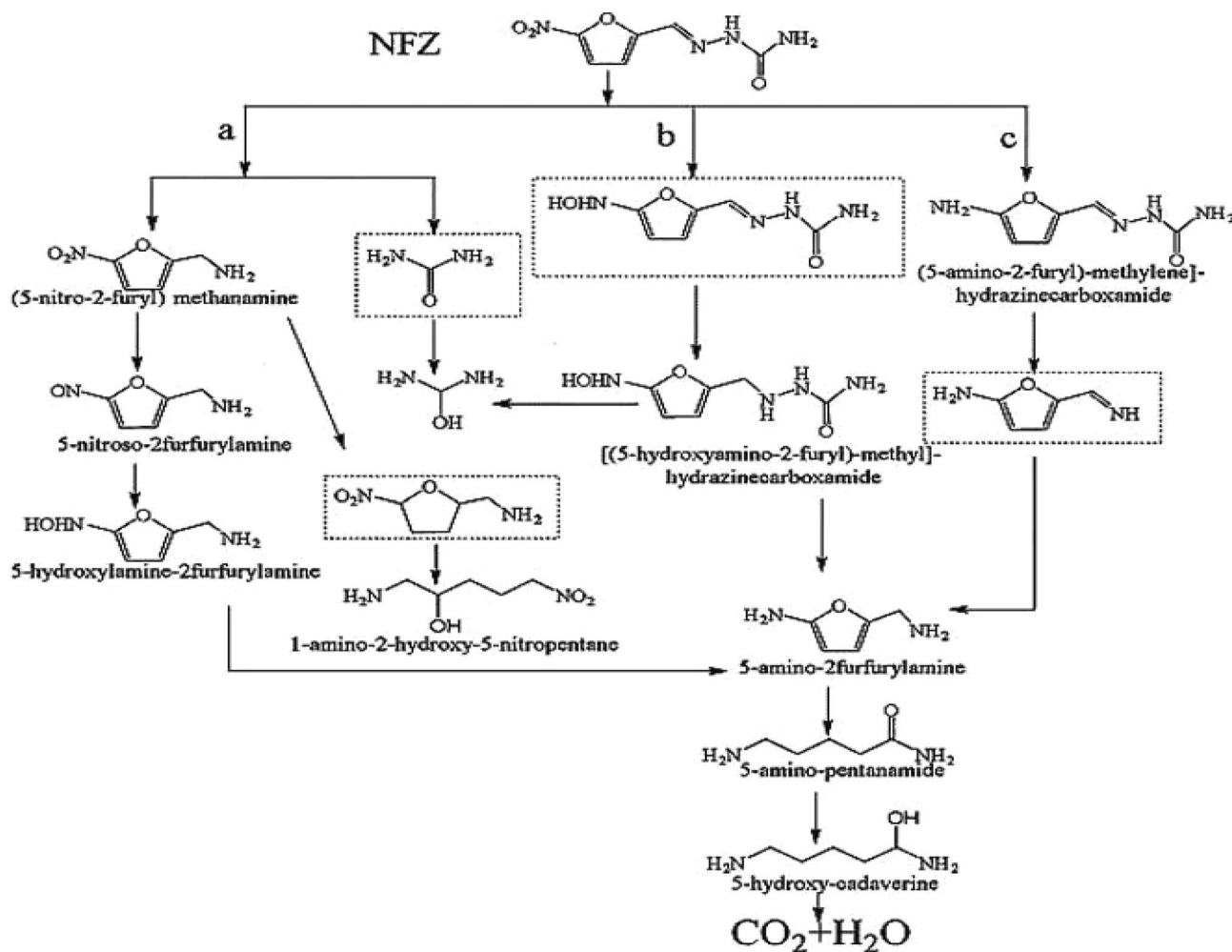
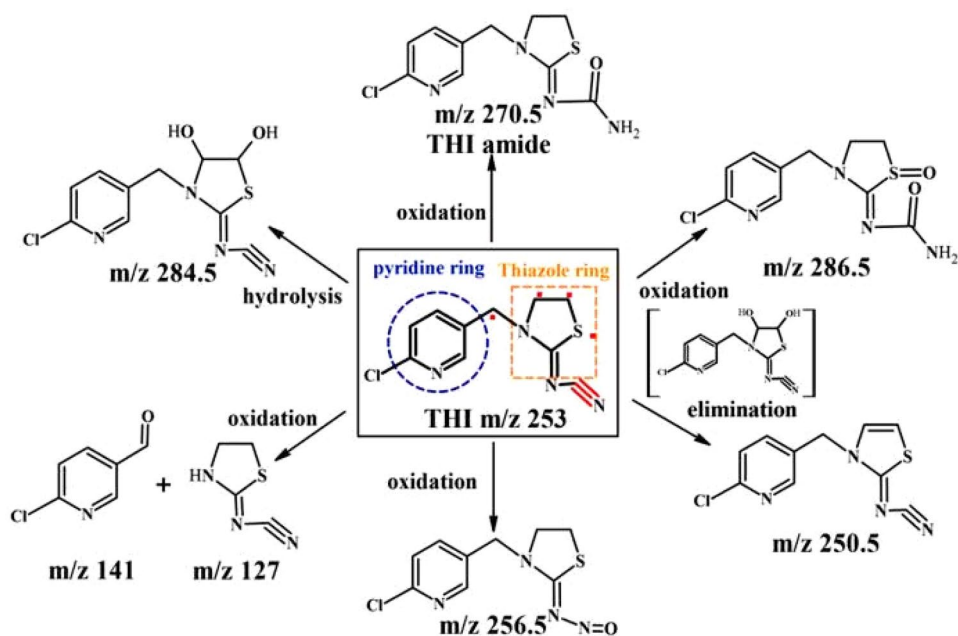


Fig. 4 Possible pathways involved in nitrofurazone (Hou et al. 2020b)

Fig. 5 Proposed degradation pathways for thiacloprid (Deng et al. 2022a, b)



This oxygen is essential as an electron acceptor during the oxidation–reduction process and electron transfer chain, where it combines with electrons released from the degradation of the substrate, such as thiacloprid at the anode. This process has enhanced the bioreactor efficiency and electricity production through the electron transfer chain. In addition, the metabolic activity of algae can degrade thiacloprid as a substrate and diversify the microbial culture (e.g., bacteria), thereby effectively increasing wastewater treatment efficiency. Additionally, there is a synergistic effect between bacteria, which degrade organic matter, and algae, which produce oxygen and nutrients in this bioreactor. This synergy contributes to electricity production and the removal of thiacloprid (Deng et al. 2022b). Another study has depicted four probable pathways for the degradation of the herbicide ametryn, based on the detection of by-products during the bio-electro-Fenton system under solar light. In this process, the hydroxyl ($\cdot\text{OH}$) radicals produced at the cathode can decompose ametryn into smaller, less toxic compounds. In addition, solar light can facilitate the oxidation of ametryn by generating holes at the anode. Furthermore, microbial metabolism can contribute to the degradation of intermediates formed in the bioreactor (Fig. 6) (Cai et al. 2023). The degradation pathways of compounds such as pesticides can be more complex than those of other compounds; however, this complexity depends on the molecular structure of the substances involved. Yi et al. (2022) present several reactor schematics in photo-bioelectrochemical systems (PBESs). Photoelectrochemical BESs utilize semiconductor materials such as titanium dioxide (TiO_2), tungsten trioxide (WO_3), and iron(III) oxide (Fe_2O_3) as electrodes. When illuminated with light, these semiconductors produce electron–hole pairs

that drive useful electrochemical reactions. For instance, a photoactive cathode can reduce oxygen gas or carbon dioxide, while the anode is powered by microbes that oxidize organic compounds. Additionally, photosynthetic BESs employ microalgae or bacteria as biocatalysts in the cathode compartment of photo-BESs. These microbes generate electrons through photosynthesis, which can then be used to reduce oxygen gas or carbon dioxide. Meanwhile, the anode is powered by the oxidation of organic compounds carried out by a different set of microbes (Yi et al. 2022). In general, the synergistic effect of photocatalysts and BESs in PBESs is illustrated in Fig. 7.

Bioenergy production

Additionally, photo-bioelectrochemical systems (PBESs) have emerged as a promising technology for bioenergy recovery by leveraging the synergistic effects of microbial catalysis and photochemical processes. These systems enhance the degradation of pollutants while generating renewable energy through bioelectricity and valuable by-products. In PBESs, the core principle involves coupling semiconductor photoelectrodes with anodes, where microorganisms act as electrocatalysts. Studies have shown that these systems can effectively reduce chemical oxygen demand (COD) by over 90% while simultaneously generating bioelectricity, with power densities reaching up to 24.4 mW/m^2 under optimal light conditions (Yi et al. 2022; Rafieenia et al. 2022). In various bioreactors, such as CW-BESs, bioenergy is primarily generated through the activities of electroactive bacteria found in the anode chamber. The process begins with the anaerobic digestion

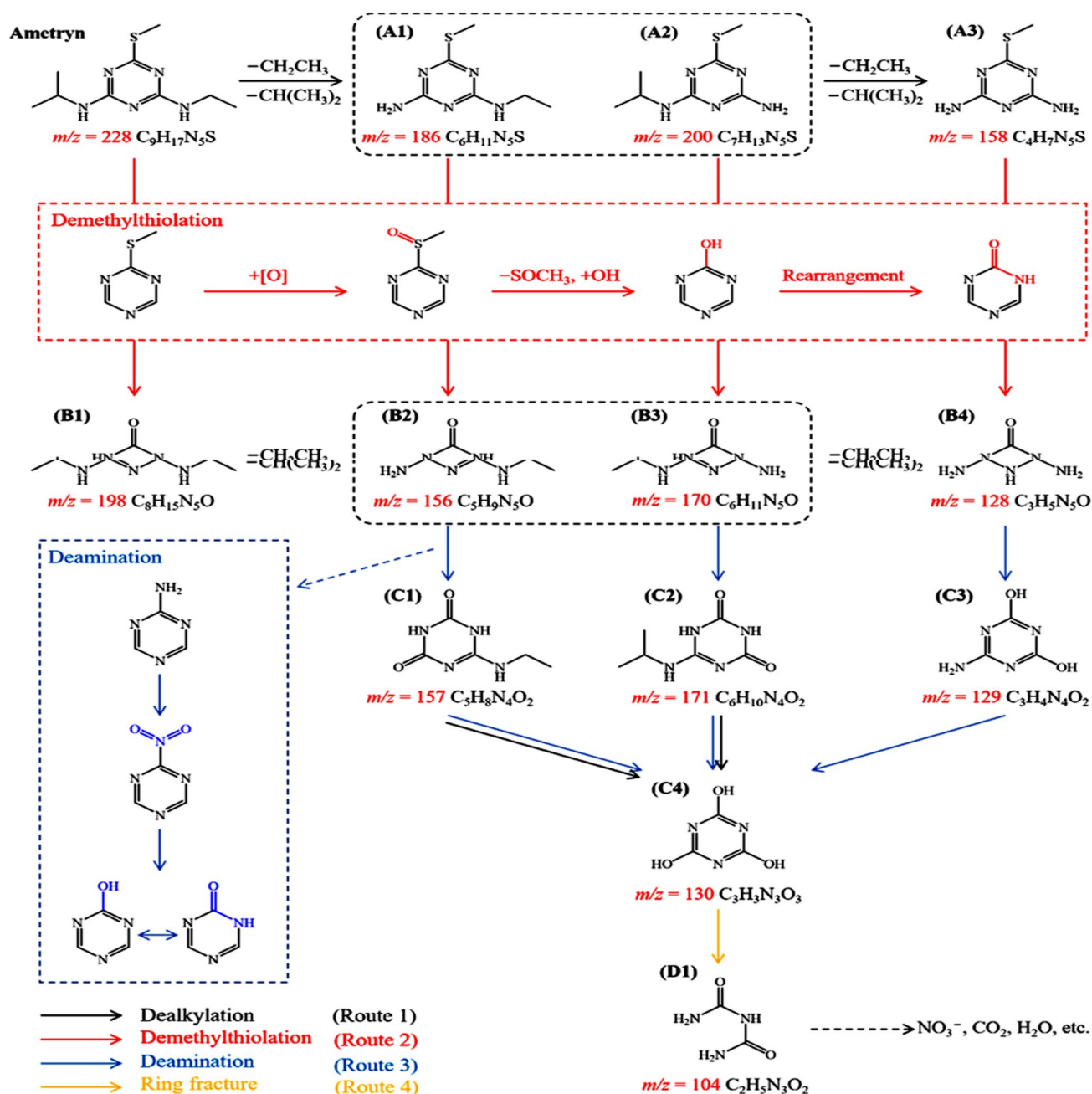
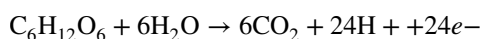


Fig. 6 Schematic representation of ametryn degradation pathways (Cai et al. 2023)

of organic materials, during which bacteria decompose substances such as sewage sludge and biochar, resulting in the release of electrons and protons. The overall reaction can be summarized as follows:



These electrons are conveyed to the photocathode through an external circuit, where they can reduce terminal electron acceptors such as oxygen or nitrates, thereby

producing electrical energy. Researchers have reported that the efficiency of this process is influenced by several factors, including the type of substrate, the composition of the microbial community, and operational conditions such as hydraulic retention time (HRT) and organic loading rate (OLR). Recent studies have highlighted significant advancements in the performance of CW-BESs. For instance, maximum power densities reported range from 0.15 mW/m² to over 600 mW/m², depending on system configurations and operational parameters. Integrating

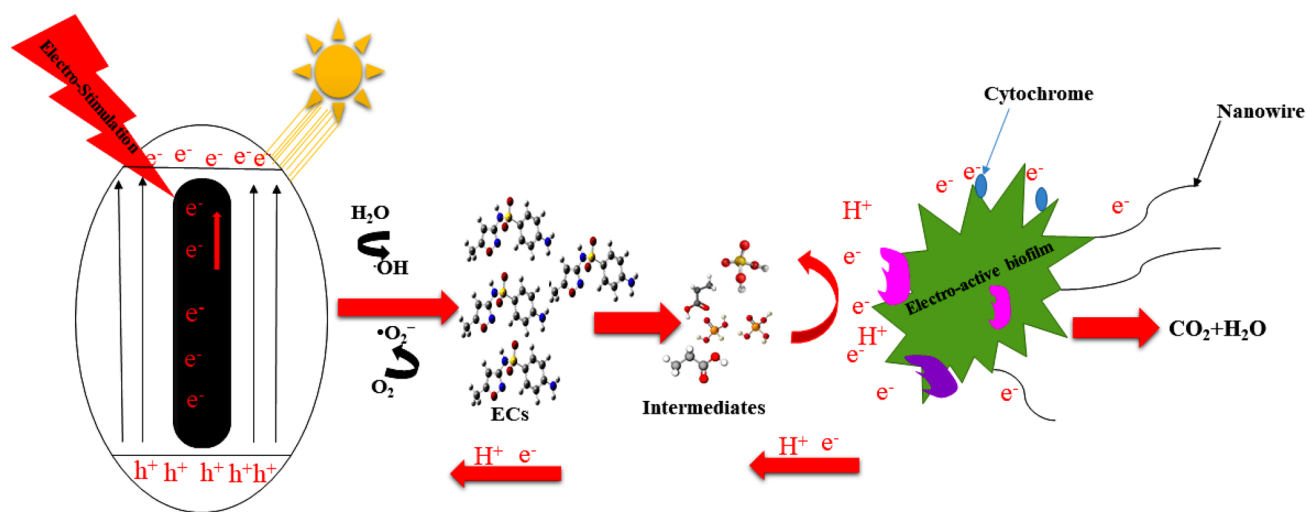


Fig. 7 Synergistic mechanism of photocatalysts and bioelectrochemical systems (BESs) in PBESs (Rafieenia et al. 2022)

biochar not only provides a carbon source, but also increases the surface area available for microbial colonization, which in turn improves the electron transfer rates and overall energy recovery (Zhang et al. 2024).

In addition, the interaction between the BES and algal or photosynthetic bacteria creates a symbiotic environment where both systems reinforce each other's performance, leading to improved energy recovery and contaminant remediation. As cited before, an algal bioreactor harnesses sunlight to promote photosynthesis, where algae absorb carbon dioxide and nutrients while releasing oxygen. This oxygen is utilized as an electron acceptor at the cathode, enhancing the electrochemical reactions within the BES. Regarding bioenergy production, the system has recorded a maximum power density of 2.2 W/m³ during 1 year of operation. The efficiency of bioenergy production in such systems can be influenced by three factors such as adequate light conditions, dissolved oxygen (DO), and retention time, which can be suitable for enhancing photosynthesis and algal biomass production (Xiao et al. 2012).

Another study, by Vinayak et al. (2021), integrated the diatom BESs with photocatalytic and photoelectric systems. In this study, diatoms act as both a source of biomass and a catalyst for electron transfer. Regarding energy output, these systems can produce power densities ranging from 20 to 30 mW/m², depending on operational conditions such as light intensity and substrate concentration in the bioreactor (Vinayak et al. 2021). Moreover, using sediment bioelectrochemical systems (BESs) with solar-enhanced bio-electro-Fenton (BEF) processes presents a novel approach to bioenergy recovery. The maximum power density achieved in this system was reported to be 44.6 W/m³, indicating a robust capacity for bioelectricity generation alongside pesticide remediation. The sediment BESs operate by harnessing the

metabolic processes of sediment-dwelling microorganisms. According to this study, several factors influence the system's performance, including adequate solar light for maximizing photocatalytic activity and enhancing hydroxyl radical generation; the composition and metabolic activities of sediment microorganisms, which affect substrate oxidation rates and electron transfer chain efficiency; and the cathode material, such as γ -FeOOH-coated carbon felt, which plays a significant role in facilitating efficient electrochemical reactions (Cai et al. 2023).

Conclusions and future perspective

PBESs, which integrate BESs with photocatalysts or photochemical processes within a single reactor, represent a promising and innovative approach. These systems enhance extracellular electron transfer (EET) by increasing the production of reactive oxidative species at photoelectrodes and facilitating interactions between microbial metabolic and enzymatic activity and the photoelectrodes. This integration subsequently improves the removal efficiency of ECs, even at low retention time, while enabling the production of electricity, biofuels, and other valuable products. Crucial and sensitive operating conditions—such as oxygen level, temperature, pH, substrate type and concentration, EC, applied voltage, and exceptionally light sources (e.g., solar, LED, fluorescent) and semiconductor photocatalysts used—play significant roles in determining the reaction mechanisms, oxidative and reductive reactions, as well as the growth, activity, structure, permeability, and microbial community diversity of electro-photo-inducing microorganisms. Therefore, optimizing these conditions is essential; for instance, selecting appropriate conductive photocatalytic materials and maintaining a stable, preferably neutral pH is critical to

ensuring the efficient operation and maintenance of PBESs. While PBESs show great potential for treating ECs, further research is needed to address other recalcitrant and hazardous compounds. Additionally, employing mathematical modeling and theoretical approaches can help elucidate the degradation mechanisms and improve the performance of PBESs in EC treatment.

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Authors contributions All authors contributed to the study conceptualization and methodology. Also, Z. Moghiseh performed the investigation and formal analysis of the research and wrote the manuscript. All authors read and approved the final manuscript.

Data availability This article is a review and does not contain any original experimental data. All data and information referenced in this review are available from the cited sources.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval and consent to participate This research did not involve any human participants and/or animals.

Consent for publication The authors consent to the publication of this work.

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